Differentiable pull-in solver

solves the saddle-node bifurcation directly · exact gradients, no surrogate fitting

$R(Y, \Lambda) = 0 \qquad J(Y, \Lambda)\, v = 0 \qquad \|v\| = 1$

Inverse design

- **22.86 V → 13.60 V (−41%)**
  electrode shape alone
- **No new mask or process step**
  same fabrication flow
- **15 s on a laptop CPU**
  no commercial licence

Amortized network

- **0.04 ms per design**
  spec → geometry, one pass
- **0 stored simulations**
  trained through the solver
- **0.56% spec error**
  hard-constraint layer

Safe-operation RL

- **+12.8% usable travel**
  from tolerance margin
- **0.0% devices destroyed**
  1,536 test episodes
- **Learns each device's ceiling**
  which cannot be measured

No training dataset. No commercial licence.

validated against 8 independent sources, 1967–2025 · exact gradient verified to 0.00%